

Example 9.3

Suppose we fit

$$\phi_0(B) X_t = \theta_0(B) Z'_t \quad (1)$$

but the residuals do not look like white noise.

Instead, looking at the ^{sample} ACF/PACF plots for the residuals $\{Z'_t\}$ against the lag, and we think that

$$\phi^*(B) Z'_t = \theta^*(B) Z_t \quad (2)$$

Take (1) and note that

$$Z'_t = \theta_0(B)^{-1} \phi_0(B) X_t.$$

Then substitute the above into (2)

$$\Rightarrow \phi^*(B) [\theta_0(B)^{-1} \phi_0(B)] X_t = \theta^*(B) Z_t$$

$$\Rightarrow \underbrace{[\theta_0(B)^{-1} \phi_0(B)]}_{\alpha(B)} X_t = \underbrace{\phi^*(B)^{-1} \theta^*(B)}_{\psi(B)} Z_t$$